



TEMASEK JUNIOR COLLEGE, SINGAPORE
Preliminary Examinations
Higher 2

MATHEMATICS

9740/02

Paper 2

15 September 2008

Additional Materials: Answer Paper
 List of Formula (MF15)

3 hours

READ THESE INSTRUCTIONS FIRST

Write your *Civics Group* and *Name* on all the work that you hand in.

Write in dark blue or black pen on both sides of the paper.

You may use a soft pencil for any diagrams or graphs.

Do not use staples, paper clips, highlighters, glue or correction fluid.

Answer **all** questions.

Give non-exact numerical answers correct to 3 significant figures, or 1 decimal place in the case of angles in degrees, unless a different level of accuracy is specified in the question.

You are expected to use a graphic calculator.

Unsupported answers from a graphic calculator are allowed unless a question specifically states otherwise.

Where unsupported answers from a graphic calculator are not allowed in a question, you are required to present the mathematical steps using mathematical notations and not calculator commands.

You are reminded of the need for clear presentation in your answers.

At the end of the examination, fasten all your work securely together.

The number of marks is given in brackets [] at the end of each question or part question.

This document consists of 5 printed pages and 1 blank page.



TEMASEK JUNIOR COLLEGE, SINGAPORE
PASSION PURPOSE DRIVE



Section A: Pure Mathematics [40 marks]

- 1 A curve C is defined parametrically by the equations

$$x = t^2 - 5 \quad \text{and} \quad y = \frac{1}{t-2} + 5, \quad t \neq 2.$$

- (i) Find $\frac{dy}{dx}$ in terms of t . [2]
- (ii) Find the equation of the normal to C at $P(4,6)$. [3]
- (iii) Find the other values of t at the points where the normal at P cuts C . [3]

- 2 (i) Express $\frac{1}{p(1 - 0.005p)}$ in partial fractions. [2]

The population growth of a certain species of animals follow the differential equation

$$\frac{dp}{dt} = ap(1 - 0.005p), \quad a \in \mathbb{R},$$

where t is the time in years and p represents the population size of the animals (in thousands).

The initial population is 150 thousand; after 5 years, the population has increased to

$$\frac{600e}{1+3e} \text{ thousands.}$$

- (ii) Show that p can be represented by the equation $p = \frac{600e^{0.2t}}{1+3e^{0.2t}}$. [5]
- (iii) After a long period of time, the population size of this species of animal will approach a particular number k . Find k . [2]
- 3 (i) Find the real numbers a and b , such that $z = -2 - 2i$ is a root of the equation $z^2 + az + b = 0$. [2]
- (ii) Solve the equation $w^3 = -2 - 2i$, giving your answers in exact form. [4]
- (iii) Using the results in (i) and (ii), deduce the roots of the equation $x^6 + 4x^3 + 8 = 0$ in the form $re^{i\theta}$, where r and θ are exact values to be determined. [3]

- 4 (a) The plane Π has equation $\mathbf{r} \cdot \begin{pmatrix} 1 \\ 2 \\ -1 \end{pmatrix} = 4$ and the line l has equation $\mathbf{r} = \begin{pmatrix} 1 \\ 1 \\ 3 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ 0 \\ -2 \end{pmatrix}$.
- (i) Find the angle between l and Π . [3]
- (ii) Find the point of intersection between l and Π . [3]
- (iii) T and J are two points on l such that distance of Π from T is equal to the distance of Π from J . Given that T has position vector $\begin{pmatrix} 1 \\ 1 \\ 3 \end{pmatrix}$, find the position vector of J . [3]
- (b) The planes Π_1, Π_2, Π_3 have equations $\mathbf{r} \cdot \begin{pmatrix} 1 \\ -1 \\ 1 \end{pmatrix} = 2$, $\mathbf{r} \cdot \begin{pmatrix} 4 \\ -2 \\ 1 \end{pmatrix} = 3$ and $\mathbf{r} \cdot \begin{pmatrix} -3 \\ -1 \\ 3 \end{pmatrix} = 4$ respectively.
- Show that the planes Π_1, Π_2 and Π_3 intersect at infinitely many points. Hence, determine the position vector of the common point for which the z-coordinate is 3. [5]

Section B: Statistics [60 marks]

- 5 Farmers Alex, Ben and Charlie are discussing their plans for a piece of land which is divided into 9 plots as numbered in the diagram below.

1	2	3
4	5	6
7	8	9

- (i) Farmer Alex wishes to plant wheat in 2 even-numbered plots and corn in 3 odd-numbered plots. How many ways are there for him to do so? [2]
- (ii) A cluster is a set of 4 adjacent plots which form a square. For example, plots 1, 2, 4 and 5 form a cluster. Farmer Ben wishes to plant 4 different types of fruits in a cluster, each in a different plot. How many ways are there for him to do so? [2]
- (iii) Farmer Charlie wishes to select 4 non-adjacent plots in which to plant 4 different types of vegetables, each in a different plot. For example, plots 1 and 2 are adjacent, but plots 1 and 5 are not. How many ways are there for him to do so? [2]

- 6 In a university, 60% of alumni members enjoyed their course of study. 70% of alumni members found jobs which they enjoyed given that they enjoyed their course of study, while 30% of alumni members did not enjoy their course of study and also found jobs which they did not enjoy. Find the exact probability that:
- (i) an alumni member found a job which he enjoyed, [2]
 - (ii) an alumni member found a job which he did not enjoy given that he did not enjoy his course of study, [2]
 - (iii) an alumni member did not enjoy his course of study given that he found a job which he enjoyed. [3]
- 7 A carpet shop in the remote town of Kikiboo receives an average of λ customers per day.
- Given that it is equally likely that the shop will receive either 3 or 4 customers, show that $\lambda = 4$. [2]
- Show that the probability of the number of visitors to the shop exceeding 30 in a given week of 7 days is 0.30965, correct to 5 significant figures. [2]
- Assuming that there are 52 weeks in a year, use a suitable approximation to find the probability that there are at least 8 but not more than 12 weeks during which this happens. [4]
- 8 In a certain country, the heights of male citizens follow a Normal distribution with mean 170 cm. If 90% of the males are taller than 165 cm, determine the standard deviation of the height of males in this country. [3]
- The heights of women in the country follow a Normal distribution with mean 162 cm and standard deviation 9.65 cm. Find the probability that a randomly chosen male of this country has a difference in height of at least 6 cm compared to a randomly chosen female of the same country. [3]
- 10 couples, each consisting of a male and female citizen, sign up for a ballroom dancing class. Find the probability that at most 4 couples have heights differing by less than 6cm. [2]
- 9 A manufacturer produces candles with a mean burning time of μ minutes and standard deviation σ minutes. From past records, the mean burning time of the candles is reported to be 28 minutes.
- (a) The burning time of 15 randomly chosen candles has a mean of 29.4 minutes and a standard deviation of 2.38 minutes. Find the smallest integer α such that there is sufficient evidence at $\alpha\%$ level of significance to say that the mean burning time has increased. State any assumptions you have to make in order to carry out the test. [5]
 - (b) Given that σ is now known to be 3.2 minutes, find the range of values of the sample mean obtained from a sample of size 100 that would lead to the rejection of the null hypothesis in favour of the alternative hypothesis that the mean burning time has changed at 2 % level of significance. [4]

- 10** A secondary school has a student population consisting of 600 boys and 720 girls. A study on the amount of time spent reading newspapers in a week is to be conducted by randomly choosing 60 male students and 72 female students from the class lists and obtaining necessary information from them.

(i) Name the sampling method used in this study, and state an advantage of this method. [2]

The data (in hours) collected from the 60 boys (denoted by x) and the 72 girls (denoted by y) is summarised as follows:

$$\sum x = 227.4, \quad \sum x^2 = 997.78, \quad \sum y = 273.6, \quad \sum (y - \bar{y})^2 = 209.35.$$

(ii) Find unbiased estimates of the means and variances of x and y . [4]

(iii) Find the probability that, on average, male students in the school spend more time reading newspapers in a week than female students in the school. [4]

- 11** With the aid of a suitable diagram, explain what is meant by the “least squares regression line of y on x ”, where x and y are two observed variable quantities of a given population. [2]

An open bottle containing 70 ml of perfume was placed in a bathroom. The amount of perfume lost to evaporation over time was recorded as follows:

No. of weeks (t):	10	20	30	40	50	60	70	80	90
Volume lost, in ml (v)	15	26	45	52	k	61	64	66	67

(i) Given that the estimated regression line of v on t is

$$v = 19.19 + 0.625t,$$
find the value of k , giving your answer to the nearest integer. Calculate the value of the product-moment correlation coefficient. [3]

(ii) If V is the value obtained by substituting a sample value of t into the equation of the regression line of v on t , find $\sum (v - V)^2$. [1]

(iii) Find the expected value of v when $t = 90$ and comment on the result obtained. [2]

(iv) In an attempt to better model the relationship between v and t , three alternatives are suggested:

(I) $v = a + bt + ct^2$, with $c < 0$

(II) $v = a + \frac{b}{t}$, with $b < 0$

(III) $v = a + be^t$, with $b > 0$

State, with a reason, which model is most appropriate and calculate the corresponding least squares regression line by performing a suitable transformation. [2]

(v) Using the model chosen in (iv), estimate the value of v when $t = 90$. Compare and comment on the result obtained here, and in (iii). [2]

END OF PAPER

